

HYDRODYNAMIC QUANTUM ANALOGS

HQA 2026

Cambridge, Massachusetts



MIT, August 10–11, 2026

MASSACHUSETTS INSTITUTE OF TECHNOLOGY

TIME	SPEAKER	TITLE
9 – 9:45am	John Bush (MIT)	Hydrodynamic quantum analogs
9:45 – 10:30	Pedro Sáenz (UNC)	Wave-particle active matter
<i>Coffee Break · 10:30 – 10:45</i>		
10:45 – 11:15	Ana Maria Cetto (UNAM)	Stochastic electrodynamics: a realist approach to revealing the physics behind quantum phenomena
11:15 – 11:45	Bauyrzhan Primkulov (Yale)	Pilot-wave dynamics beyond resonance
11:45 – 12:15	André Nachbin (WPI)	2 HQAs: wave-droplet oscillators and quantum graphs
LUNCH · 12:15 – 1PM		
1 – 1:30	Paul Milewski (PSU)	Strong nonlinear resonances of water waves in confined domains
1:30 – 1:45	Daniel Harris (Brown)	Drop-bath impacts
1:45 – 2	Larry Wilen (Yale)	Manipulating bouncing droplets with tunable global forcing
2 – 2:15	Saiful Tamim (UNC)	Wave-mediated walking of solid particles at a fluid interface
2:15 – 2:30	Johnathan Hoggarth (Yale)	Drop generation via stretching of liquid ligaments
2:30 – 2:45	James Day (MIT)	Ensembles of walking droplets
2:45 – 3:15	Steven Rodriguez (ONR)	Leveraging hydrodynamic analogs in optical beam transport
3:15 – 3:45	Georgi Gary Rozenman (MIT)	HQAs of Aharonov–Bohm, Kennard phase and triple-slit diffraction
<i>Coffee Break · 3:45 – 4</i>		
4 – 4:30	Arnaud Lazarus (MIT)	Engineering quantization of waves in periodic media: towards GR analogs in HQA
4:30 – 5:00	Joel Been (MIT)	Analogizing exciton-polaritons with walker wakes
5:00 – 5:30	Austin Blitstein (UNC)	Spatiotemporally nonlocal forces in pilot-wave hydrodynamics
5:30 – 5:45	Xinyun Liu (UNC)	Collective pilot-wave dynamics: experiments
5:45 – 6:00	Ian Stevenson (UNC)	Collective pilot-wave dynamics: simulations and theory
<i>Break — Zoom Social · 6:00 – 6:45pm</i>		
DINNER AT PAGU · 7PM		

TIME	SPEAKER	TITLE
9 – 9:30am	Matt Durey (Glasgow)	Orbital dynamics and energetics of wave-propelled particles
9:30 – 10	Rahil Valani (Oxford)	Dynamical origins of HQAs in Lorenz-like models
10 – 10:30	Valeri Frumkin (BU)	From topological transport to emergent localization in HQAs
<i>Coffee Break · 10:30 – 10:45</i>		
10:45 – 11:15	Giuseppe Pucci (Calabria)	Single-particle diffraction with a hydrodynamic pilot-wave model
11:15 – 11:30	Anand Oza (NJIT)	Theoretical results on stochastic pilot-wave dynamics
11:30 – 11:45	Ludovico Giorgini (MIT)	Stochastic pilot-wave dynamics
11:45 – 12pm	Frane Ljubetic (UNC)	Influence of path memory on the stochastic transport of walkers
LUNCH · 12 – 1PM		
1 – 1:15	Yves Desille (Lille)	Acoustic pilot-wave dynamics
1:15 – 1:30	Zahra Shamsi (Yale)	Pilot-wave hydrodynamics on noisy surfaces
1:30 – 1:45	Diego Chavez (MIT)	Pilot-wave dynamics in 3D
1:45 – 2:00	Andras Bencze (BU)	Non-Markovian branched flow in pilot-wave hydrodynamics
2:00 – 2:30	Dave Darrow (MIT)	A Lorentz-covariant Lagrangian framework for pilot-wave dynamics
2:30 – 2:45	Haoyi Wang (UNC)	The oscillatory Cheerios effect of bouncing droplets
2:45 – 3	Christopher Griffin (PSU)	Quantum mechanics on Riemannian manifolds
3 – 3:30	Adam Kay (MIT Alum)	What Bell's theorem really means
3:30 – 3:45	Kosta Papatryfonos (Lille)	Static Bell tests with walkers
3:45 – 4:15	Alvaro Lopez (Madrid)	Bell tests with a Lorenz model of walkers
CONCLUDING REMARKS · 4:15 – 5		



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